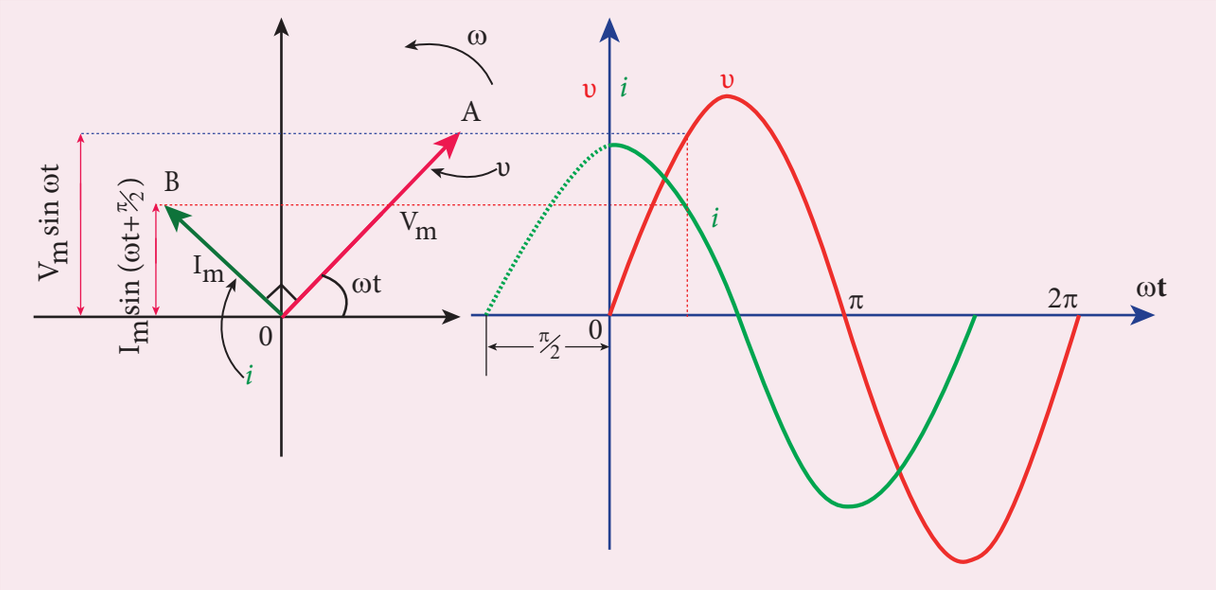


Figure 4.50 Phasor diagram and wave diagram for AC circuit with C

EXAMPLE 4.20

A 400 mH coil of negligible resistance is connected to an AC circuit in which an effective current of 6 mA is flowing. Find out the voltage across the coil if the frequency is 1000 Hz.

Solution

L = 400 x 10^{-3} H; I_{eff} = 6 x 10^{-3}A

f = 1000 Hz

Inductive reactance, $X_L = L\omega = L \times 2\pi f$

$= 2 \times 3.14 \times 1000 \times 0.4$

$= 2512\ \Omega$

Voltage across L,

$V = I X_L = 6 \times 10^{-3} \times\ 2512$

$V = 15.072 V (RMS)$

EXAMPLE 4.21

A capacitor of capacitance $\frac{10^2}{\pi}\mu F$ is connected across a 220 V, 50 Hz A.C. mains. Calculate the capacitive reactance, RMS value of current and write down the equations of voltage and current.

Solution

$C = \frac{10^2}{\pi} \times 10^{-6} F,\ V_{RMS} = 220\ V; f = 50 Hz$

(i) Capacitive reactance,

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

$$= \frac{1}{2 \times \pi \times 50 \times \frac{10^{-4}}{\pi}} \quad = 100\,\Omega$$

(ii) RMS value of current,

$$I_{RMS} = \frac{V_{RMS}}{X_C} = \frac{220}{100} = 2.2\ A$$

(iii) $V_m = 220 \times \sqrt{2} = 311\ V$

UNIT-4(XII-Physics_Vol-1).indd 258 04-03-2019 10:41:02